ECON 3201
Econometrics for Economics and Finance

Assignment 1
Due Thursday October 8, 2026

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Instructions

Answer all questions. Please complete your assignment using Quarto. Submit both your Quarto file and pdf output file.

Theory Questions

Q1

Create a personal Quarto website hosted on GitHup pages. For reference, please see the slides for **website** on Nexus. Your website should include the following:

i. An "Index" or "Home" page. This page should include the following:

- Your name
- Your photo (optional)
- Your credentials
- An "About Me" section that provides a brief (one paragraph) introduction to yourself. [10 Marks]

ii. A blog page. This will be the landing site for future assignments [5 Marks]
iii. Create a blog post and upload the finished PDF for this assignment. [3 Marks]
iv. Navigation bar with "Home" and "Blog" buttons. The navigation bar is edited in the "_quarto.yml" file. [2 Marks]

> http://localhost:3336/home.html

Q2

Write your solutions using LaTeX. Show your work. You may use R to compute the answers.

Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12$ be a uniformly distributed variable with probability distribution function $f_X(x) = 1/12$.

(a) Check that $f_X(x)$ is a true probability distribution.

The distribution is valid because all probabilities are non-negative and the probabilities add up to 1.

$$f_X(x) \geq 0$$

and

$$\sum_{x=1} f_X(x) = 1.$$

Since

$$f_X(x) = \frac{1}{12} > 0$$

for all $x = 1, 2, \ldots, 12$, and

$$\sum_{x=1}^{12} f_X(x) = \sum_{x=1}^{12} \frac{1}{12} = 12\left(\frac{1}{12}\right) = 1,$$

$f_X(x)$ is a true probability distribution.

(b) Compute the expected value of X.

$$E[X] = \sum_{x=1}^{12} x f_X(x).$$

Therefore,

$$\begin{aligned}
E[X] &= \sum_{x=1}^{12} x\left(\frac{1}{12}\right) \\
&= \frac{1}{12}\sum_{x=1}^{12} x \\
&= \frac{1}{12}(1+2+3+4+5+6+7+8+9+10+11+12) \\
&= \frac{1}{12}(78) \\
&= 6.5.
\end{aligned}$$

(c) Compute the variance of X.

The variance of X is

$$Var(X) = E[X^2] - [E[X]]^2.$$

First, compute $E[X^2]$:

$$\begin{aligned}
E[X^2] &= \sum_{x=1}^{12} x^2 f_X(x) \\
&= \sum_{x=1}^{12} x^2\left(\frac{1}{12}\right) \\
&= \frac{1}{12}(1^2 + 2^2 + \cdots + 12^2) \\
&= \frac{1}{12}(650) \\
&= \frac{325}{6}.
\end{aligned}$$

Therefore,

$$\begin{aligned}
Var(X) &= E[X^2] - [E[X]]^2 \\
&= \frac{325}{6} - (6.5)^2 \\
&= \frac{325}{6} - 42.25 \\
&= 11.9167.
\end{aligned}$$

Thus,

$$Var(X) \approx 11.9167.$$

Q3

Let $X \sim N(4, 5)$. Find the probability of each event:

(a) $Pr(X \leq 6)$

$$Z = \frac{X - \mu}{\sigma}.$$

Since $\mu = 4$ and $\sigma = \sqrt{5}$,

$$P(X \leq 6) = P\left(Z \leq \frac{6-4}{\sqrt{5}}\right) = P(Z \leq 0.8944).$$

Using the standard normal distribution,

$$P(Z \leq 0.8944) \approx 0.8145.$$

Therefore,

$$P(X \leq 6) \approx 0.8145.$$

(b) $Pr(X > 4)$

$$P(X > 4) = P\left(Z > \frac{4-4}{\sqrt{5}}\right) = P(Z > 0).$$

By symmetry of the standard normal distribution,

$$P(Z > 0) = 0.5$$

Therefore,

$$P(X > 4) = 0.5$$

(c) $Pr(-1 \leq X < 1)$

$$\begin{aligned} P(-1 \leq X < 1) &= P\left(\frac{-1-4}{\sqrt{5}} \leq Z < \frac{1-4}{\sqrt{5}}\right) \\ &= P(-2.2361 \leq Z < -1.3416) \end{aligned}$$

Therefore,

$$\begin{aligned} P(-1 \leq X < 1) &= P(Z < -1.3416) - P(Z < -2.2361) \\ &\approx 0.0899 - 0.0127 \qquad = 0.0772. \end{aligned}$$

Therefore,

$$P(-1 \leq X < 1) \approx 0.0772.$$

Q4

Let $Z \sim N(0, 1)$.

(a) Compute $Pr(Z \leq 0.83)$

$$P(Z \leq 0.83) = 0.7967.$$

Therefore,

$$P(Z \leq 0.83) = 0.7967.$$

(b) Compute $Pr(-0.83 \leq Z \leq 0.83)$

Using the symmetry of the standard normal distribution,

$$P(-0.83 \leq Z \leq 0.83) = P(Z \leq 0.83) - P(Z < -0.83).$$

From the standard normal distribution table,

$$P(Z \leq 0.83) = 0.7967$$

and

$$P(Z < -0.83) = 0.2033.$$

Therefore,

$$\begin{aligned} P(-0.83 \leq Z \leq 0.83) &= 0.7967 - 0.2033 \\ &= 0.5934. \end{aligned}$$

Therefore,

$$P(-0.83 \leq Z \leq 0.83) = 0.5934$$

R Excercises

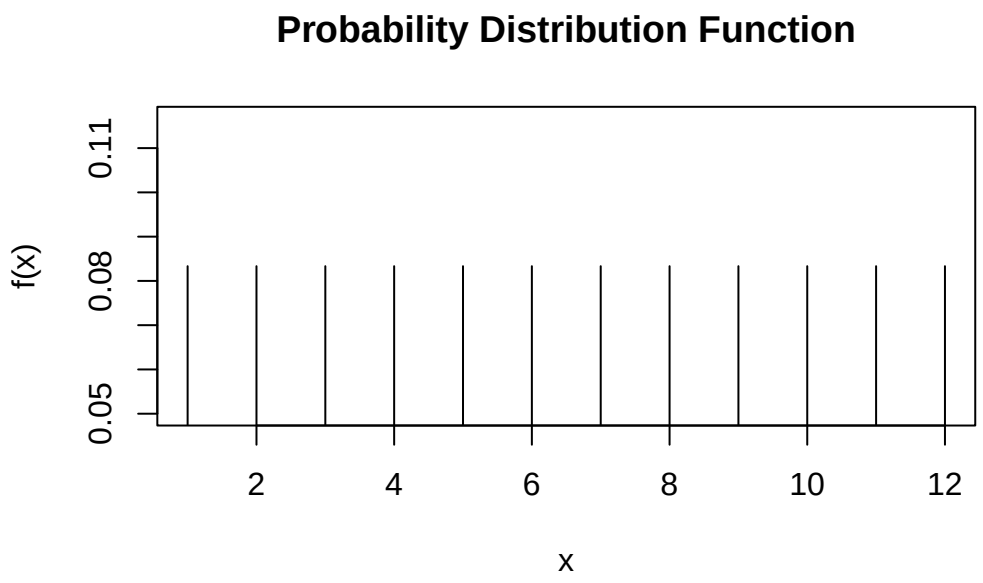
Q5

Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12$ be a uniformly distributed variable with probability distribution function $f_X(x) = 1/12$.

(a) Plot the probability distribution function $f_X(x)$ in R.

```
x= 1:12
p= rep(1/12, 12)

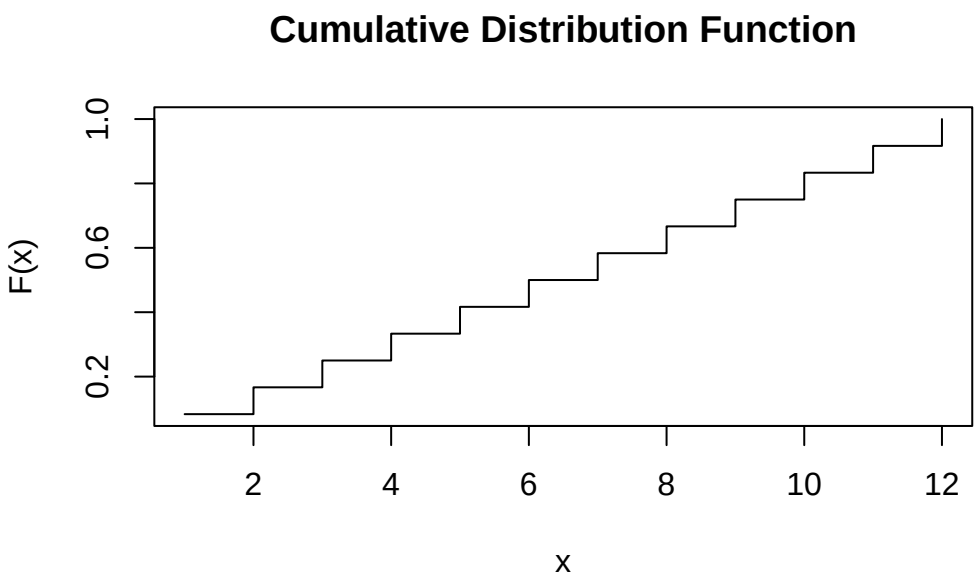
plot(x, p, type = "h", xlab = "x", ylab = "f(x)", main = "Probability Distribution Functio
```



(b) Plot the cumulative probability distribution function $f_X(x) in R.$

```
cdf= cumsum(p)

plot(x, cdf, type = "s", xlab = "x", ylab = "F(x)", main = "Cumulative Distribution Functi
```



(c) Compute the expected value of X in R.

```
expected_value= sum(x * p)
expected_value
```

```
[1] 6.5
```

(d) Compute the variance of X in R.

```
variance= sum((x - expected_value)^2 * p)
variance
```

```
[1] 11.91667
```

Q6

The standard normal distribution function is given by

$$f(x) = \frac{e^{-(x^2/2)}}{\sqrt{2\pi}\sigma}$$

(a) Define this function in R. (Note: π is coded as `pi` and e is coded as `exp`.

```
f= function(x) {(1 / sqrt(2 * pi)) * exp(-x^2 / 2)}
f
```

```
function (x)
{
    (1/sqrt(2 * pi)) * exp(-x^2/2)
}
```

(b) Integrate this function over the limit $[-\infty, \infty]$ using the `integrate()` command in R. Infinity in R is written as `Inf`. To evaluate the integral, use `integrate()$value`

```
total_prob= integrate(f, lower = -Inf, upper = Inf)$value
total_prob
```

```
[1] 1
```

(c) Compute the expected value of the standard normal distribution in R and confirm it is equal to 0.

```
expected_val_func= function(x) {x * f(x)}

expected_value= integrate(expected_val_func, lower = -Inf, upper = Inf)$value
expected_value
```

```
[1] 0
```

(d) Compute the variance of the standard normal distribution in R and confirm it is equal to 1.

```
variance_func= function(x) {(x^2) * f(x)}

variance= integrate(variance_func, lower = -Inf, upper = Inf)$value
variance
```

```
[1] 1
```

(e) Compute $Pr(Z \leq 0.83)$

```
prob_e= integrate(f, lower = -Inf, upper = 0.83)$value
prob_e
```

```
[1] 0.7967306
```

(f) Compute $Pr(-0.83 \leq Z \leq 0.83)$

```
prob_f= integrate(f, lower = -0.83, upper = 0.83)$value
prob_f
```

```
[1] 0.5934612
```

(g) Compute $Pr(-0.83 \leq Z \leq 0.83)$ using the `pnorm()` command and compare your results to (f).

```
prob_g= pnorm(0.83) - pnorm(-0.83)
prob_g
```

```
[1] 0.5934612
```

Q7

Let X denote the prison sentence, in years, for people convicted of auto theft in a particular province in Canada. Suppose the PDF of X is given by:

$$f(X) = \frac{1}{9}x^2, \ 0 < x < 3.$$

(a) Define $f(x)$ in R.

```
f= function(x) {(1 / 9) * x^2}
f
```

```
function (x)
{
    (1/9) * x^2
}
```

(b) Integrate $f(x)$ over the interval [0,3].

```
total_prob= integrate(f, lower = 0, upper = 3)$value
total_prob
```

```
[1] 1
```

(c) Compute the expected prison sentence.

The expected value is given by $E[X] = \int_0^3 x \cdot f(x)\, dx$:

```
# Function for x * f(x)
exp_func= function(x) {x * f(x)}

# Integrate x * f(x) from 0 to 3
expected_sentence= integrate(exp_func, lower = 0, upper = 3)$value
expected_sentence
```

```
[1] 2.25
```

Q8

Let X follow a Student t distribution with M degrees of freedom. Then

$$X = \frac{Z}{\sqrt{W/M}} \sim t_M,$$

where Z is a standard normal variable and W is a chi-squared variable with M degrees of freedom.

(a) On the same figure plot the following:

(i) Plot the standard normal probability distribution using `curve()`. Specify the range as [-5,5] using `xlim`. Name the `ylab` "Density". The main title should be "Densities of t Distribution".

(ii) Plot the Student t distribution with 5 degrees of freedom using `curve()` and `dt()`. Specify the range as [-5,5] using `xlim`. Choose a different colour, i.e. `col=2`. To learn more about `dt`, type `?dt` in your console.

(iii) Plot the Student t distribution with 15 degrees of freedom using `curve()` and `dt()`. Specify the range as [-5,5] using `xlim`. Choose a different colour, i.e. `col=3`. To learn more about `dt`, type `?dt` in your console.

(iv) Plot the Student t distribution with 25 degrees of freedom using `curve()` and `dt()`. Specify the range as [-5,5] using `xlim`. Choose a different colour, i.e. `col=4`. To learn more about `dt`, type `?dt` in your console.

(v) Include the following legend in you plot:
`legend("topright", c("N(0, 1)", "M=2", "M=4", "M=25"), col = 1:4, lty = c(1, 1, 1, 1))`

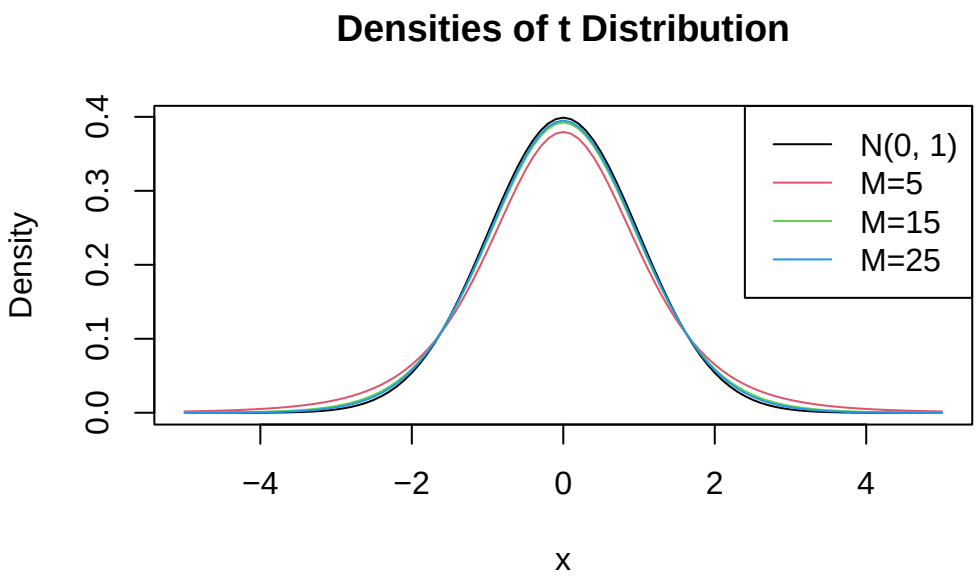
```r
# (i) Primary plot command - opens the plot window
curve(dnorm(x), xlim = c(-5, 5), ylab = "Density", main = "Densities of t Distribution", co

# (ii) Add t-distribution df = 5
curve(dt(x, df = 5), xlim = c(-5, 5), col = 2, add = TRUE)

# (iii) Add t-distribution df = 15
curve(dt(x, df = 15), xlim = c(-5, 5), col = 3, add = TRUE)

# (iv) Add t-distribution df = 25
curve(dt(x, df = 25), xlim = c(-5, 5), col = 4, add = TRUE)

# (v) Add legend
legend("topright", legend = c("N(0, 1)", "M=5", "M=15", "M=25"), col = 1:4, lty = 1)
```



(b) What do you observe as the degrees of freedom increase?

As the degrees of freedom (M) increase, the Student t-distribution approaches the standard normal distribution $N(0,1)$. Specifically, the tails become thinner, the peak becomes taller, and by $M = 25$, the t-distribution curve is identical to the standard normal curve.